Rishi Vora

BS-MS in Physics from IISER Mohali

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Education

2021 — 2026

Indian Institute of Science Education and Research (IISER), Mohali
Integrated Bachelors & Masters, with Major in Physics and Minor in Data Science

- **CPI:** 8.0 / 10
- **Relevant Courses:** Foundations of Quantum Mechanics and Quantum Information | Advanced Quantum Mechanics | Introduction to Data Science | Machine Learning and Artificial Intelligence | Computational Methods in Physics | Algorithms and Complexity | Nonlinear Dynamics and Chaos | Modelling Complex Systems

Research Experience

12/2025 — 09/2026

Quantum generative machine learning, IISER Pune
Masters Thesis

- **Guide:** Prof. Santhanam | **Code:** github.com/Vortriz/thesis
- Created a new Quantum Generative model that requires fewer resources and has significantly faster convergence compared to existing models, such as QuDDPM.
- Applied it to various tasks, such as learning many-body phases, states distributed over nontrivial topology, localized states of Quantum Kicked Rotor etc.
- Benchmarked gradient-free optimization strategies such as QNSPSA, SPSA, and Rotosolve against gradient-based approaches.
- Created a Julia implementation using Yao.jl, with optional CUDA acceleration.

Summer 2024

Reading project on QKD protocols, IISER Mohali
Summer reading project

- **Guide:** Prof. Arvind
- Reviewed working of BB84, B92 and E91 QKD protocols.
- Worked out Shor and Preskill's security proof of BB84 protocol.

Summer 2023

Quantum algorithms and circuit decomposition, IISER Mohali
Summer internship

- **Guide:** Dr. Sandeep Goyal | **Code:** github.com/Vortriz/Summer2023
- Implemented Full Adder, Multiplier, Superposition Adder, and XOR adder circuits in Qiskit.
- Attempted a circuit decomposition scheme using Solovay-Kitaev algorithm.

Skills & Certifications

2024

Qiskit Global Summer School 2024
IBM

- Certificate of Quantum Excellence
- Lectures, interactive sessions, and programming tasks on Circuit Transpilation, Modelling and Characterization of Hardware Noise, Execution on Noisy Quantum Hardware using Qiskit.

2023 | **Qiskit Global Summer School 2023**
IBM

- Certificate of Quantum Excellence
- Lectures, interactive sessions, and programming tasks on Dynamic circuits, Phase estimation, Factoring, Error mitigation, VQE, and Iterative phase estimation.

Programming & Tools | **Python** (Qiskit, PyTorch, TensorCircuit) | **Julia** (Yao.jl) | **Linux** (NixOS) | **Git** | **Typst** | Code written as a part of the coursework can be found on my website.

Spoken Languages | **English** (Fluent) | **Hindi** (Native) | **Gujarati** (Native)

Hobbies | I actively develop and contribute to FOSS projects on GitHub, create websites once in a while, and spend my downtime listening to music and watching eclectic films.

Scholarships

2022 — 2026 | **IISER Mohali** — Merit-Cum-Means Scholarship

Bi-annual scholarship supporting Bachelor's and Master's studies.

Extracurriculars

June 2026 | **Won 2 bounties in unitaryHACK 2026**
Unitary Foundation

- Annual virtual hackathon organized by Unitary Foundation for contributing to open-source quantum ecosystem.
- Bounties:
  - `quantumsavory.jl` – `show` and visualization methods for multipartite Gaussian states from Gabs.
  - `paulipropagation.jl` – Embed example Jupyter notebooks into docs website.

March 2026 | **Conducted Workshops on Python Tooling and Typst**
IISER Pune

- **Link**: github.com/Vortriz/workshops-and-presentations
- Conducted interactive workshops focused on addressing problems with Python tooling ecosystem and showcasing Typst for high-quality document preparation.

2023 — 2024 | **Convenor of Turing Club (Computation Club)**
IISER Mohali

- Organized Hackathons, Workshops, Talks etc.
- Took programming help sessions.